

Books & arts

Isaac Newton and Charles Darwin in London's Westminster Abbey.

The gushing provides little real sense of Hawking's character. Nor does it do justice to key people such as Jacob Bekenstein, whose work on entropy in black holes Hawking initially rejected and then adopted as central to his own theories. And the book does not provide any sense of the massive technological, institutional and bureaucratic infrastructure on which Hawking depended that is masterfully described in Hélène Miallet's 2012 book *Hawking Incorporated*.

Physics versus philosophy

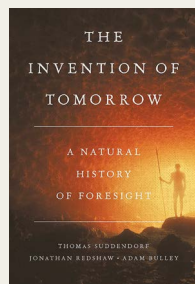
The second source of irritation is Hertog's scorn for philosophy. He treats philosophers – as Hawking did during his lifetime – as rivals who think about concepts such as whether there was a Big Bang or not, and suggests that physicists love one-upping them by applying mathematical approaches to what they think are philosophical questions. But that mistakes what philosophers do. Whereas scientists study the natural world, philosophers of science study how scientists study the natural world. Hawking and Hertog's attitude reminds me of the lament attributed to Socrates about those who think their knowledge about one thing entitles them to think they are similarly knowledgeable about many other things.

Hertog's treatment of Hannah Arendt, one of the most important philosophers of the twentieth century, is particularly ludicrous. She and Hawking each said they are against a "view from nowhere", which Hertog interprets as meaning that "Hannah meets Stephen". With this, Hertog seems to urge fellow theorists to deny the classical picture that the cosmos has "a definite spacetime with a well-defined starting point and a unique evolution". Arendt, however, criticized the theoretical and mathematical language that claims ultimate objectivity, detaches language from common sense and distances our understanding of the Universe from experience – a line of reasoning that has nothing to do with Hertog's.

By the end of the book, I couldn't help but recall a remark by the Soviet physicist Lev Landau – which Hertog quotes rather dismissively and without attribution – that cosmologists are "often in error but never in doubt". The remark seems spot on. Still, disciplinary arrogance aside, *On the Origin of Time* lets you enjoy the cosmologists, understand their theories and realize their flaws, yet sympathize with those whose confidence will soon be demolished. The book even makes you look forward to their next 'final' theory.

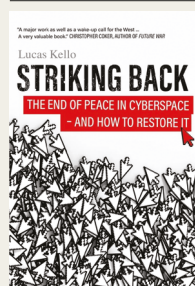
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Books in brief



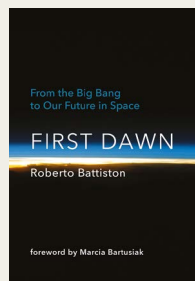
The Invention of Tomorrow

Thomas Suddendorf, Jon Redshaw & Adam Bulley *Basic* (2022)
"The human mental time machine is a complex and powerful device that allows us to imagine virtually whatever, wherever, whenever," remarks this thoughtful, far-ranging study of foresight by three cognitive scientists. Among their numerous lively examples, from Palaeolithic to modern times, are children whom the group observed responding to a reward being dropped down a tube with two exits. Aged two, they generally put a hand under only one exit, but by four most covered both – having learnt to think ahead.



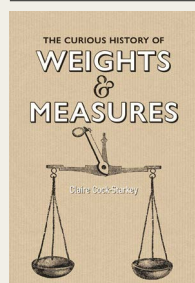
Striking Back

Lucas Kello *Yale Univ. Press* (2022)
Reports of Russian interference in the 2016 US presidential election revealed a threat to peace. Yet it went unpunished, like many political cyberattacks. This complacency must end, argues international-relations specialist Lucas Kello in his disturbing book, recommended by former directors of US and UK intelligence agencies. Western democracies are "adept at technological innovation" but "behind the times" on its security implications, he says. He suggests a "punctuated deterrence" approach that targets the interests of totalitarian nations.



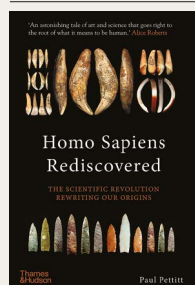
First Dawn

Roberto Battiston *MIT Press* (2022)
In 1923, Edwin Hubble's observations of the Andromeda nebula proved that there were galaxies beyond the Milky Way. Astroparticle physicist Roberto Battiston compares such revelations to humanity's changing understanding of dawn. Rather than daybreak "running to meet us", we now know that Earth's rotation moves us towards it. Battiston's elegant book on the history of knowledge about the cosmos – oddly unillustrated – examines the Sun's birth, how planets form, life's origins, black holes, gravitational waves and more.



The Curious History of Weights and Measures

Claire Cock-Starkey *Bodleian Library Publishing* (2023)
"Man is the measure of all things," said Greek philosopher Protagoras 2,500 years ago. Judging from the mind-boggling variety of body-based units, such as the cubit, that were used until measurements were standardized, "he wasn't wrong", says Claire Cock-Starkey in her enjoyable, if Anglocentric, illustrated history. She focuses on weight, length, area, volume and culinary measures, but also discusses the Beaufort wind scale, the Scoville scale of spiciness and the Bristol stool scale for monitoring bowel function.



Homo Sapiens Rediscovered

Paul Pettitt *Thames & Hudson* (2022)
Modern *Homo sapiens* owe much to our ice age ancestors, observes archaeologist Paul Pettitt after almost three decades of research. Their innovations include sewing needles, tailored clothing, jewellery, burial of the dead, art, dogs, weapons, tents, lamps, villages, and even, possibly, a form of writing. Genetic advances have transformed understanding of early human evolution, but "we are far more than our DNA", writes Pettitt. Looking at sites around the world, he rediscovers their ancient inhabitants as people. **Andrew Robinson**